ZERO TRUST DESIGN CLINIC

ZTA Design Clinic Workbook

This is your team's answer sheet. Work through one use case at a time: sketch your design, explain how it solves the customer's problems, and list the Cisco products you would use. Your proctor scores each use case out of 100 and gives you feedback.

How to use this workbook

1. **Pick your roles.** As a group, assign the four roles — NetOps, Information Security, Compliance and Scribe. The Scribe records the team's answers in this workbook.
2. **Understand the customer.** Review the current customer architecture and, for each use case, its requirements and pain points. The key points are repeated in this workbook so you can answer them directly.
3. **Design your solution.** Agree on a Cisco Zero Trust design that meets every requirement and solves each pain point. Do use cases 1–3 first (required); 4–5 are bonus if time permits.
4. **Draw your diagram.** Sketch your proposed network in the diagram box — or paste a diagram you made with the Cisco Circuit app or another AI tool.
5. **Write your responses.** In the tables, explain how your design addresses each pain point and requirement. Then list each Cisco product, what it does, and why you chose it — call out any Cisco differentiator.
6. **Present to your proctor.** Be ready to walk your SME / proctor through your design and explain your choices. They score each use case out of 100 and give you feedback.

This is a 60-minute activity. Do use cases 1–3 (required) first; 4–5 are stretch goals if you finish early. Keep **one diagram** and grow it as you go, and answer in **short bullets**.

- Use cases 1–3 are the deliverable; 4–5 are stretch goals if you finish early.
- Keep one diagram and grow it use case to use case — don't start over each time.
- Answer in short bullet points, not paragraphs. Name the product, say what it does, say why.
- Ask your proctor for a 60-second sanity check after Use Case 1 so you're on the right track.

How your work is scored. Each use case is worth **100 points** — **Mapping 40** (how well you address the pain points & requirements) · **Products 30** (the right Cisco products, explained clearly) · **Diagram 30**. Aim for the top level: **Associate** 60–74 · **Professional** 75–89 · **Expert** 90–100. New to a product name? See the **Glossary** at the end and the product primer in your Activity Document.

Team / table #: ____________________ **Members:** ____

Use Case #1 — Managed Environment; Segmentation

Foundation · Required | 100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here — or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

CUSTOMER PAIN POINT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Prior segmentation failed. Using VLANs/IP subnets is complex and tedious.	
Profiling is labor intensive and staff unable to keep up with constant new devices joining	
Failing PCI audits	
Corporate PCs not running latest AntiVirus and OS updates	
No budget for buying more firewalls to do segmentation	

STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

CUSTOMER REQUIREMENT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Employees / IoT / deskagents (Mac-minis) / Guest groups	

IoT should be restricted from the other groups but need data center and Internet access	
Guests should have Internet access only	
Employee and IoT require controlled internet, and data center application access	
Deskside Mac-minis have access to on-prem data center only	

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled — explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

CISCO PRODUCT / FEATURE	WHAT IT DOES / HOW IT WORKS	WHY YOU CHOSE IT (CISCO DIFFERENTIATOR)
ISE		
ISE-SGT		
ISE SGT ACL policy		
ISE pxGrid		
Catalyst SD-WAN		
FMC / SCC / FTD		

FMC / SCC / FTD — Source/Dest SGT policy enforcement		
Cisco Secure Workload — App Dependency Mapping		
Cisco Secure Workload — pxGrid integration with ISE		

PROCTOR USE ONLY — SCORING & FEEDBACK

CRITERION	MAX	SCORE
Pains & requirements mapping	40	
Cisco products (how & why)	30	
Diagram	30	
Total	**100**	

Overall level

- ☐ Associate `60–74`
- ☐ Professional `75–89`
- ☐ Expert `90–100`

What the team did well

Where they can improve

Use Case #2 — Remote Worker: Zero Trust Access

Foundation · Required **100 pts**

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here — or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

CUSTOMER PAIN POINT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Full client VPN has over permissive privileges. Not considered Zero Trust	
User satisfaction is low due to manual VPN connection daily	
Too many sign-ons per application. Low user satisfaction.	
Want to eliminate password logins to minimize stolen credentials	
Existing Okta IdP solution getting too expensive	
CIO wants to modernize branches with SASE	

STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

CUSTOMER REQUIREMENT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
All corporate users can use full client VPN to access selected legacy private applications. But most prefer ZTNA for modern applications.	
All corporate users need controlled internet access (block gambling and gaming)	

Contractors can only access private applications via clientless RDP access.	
Improve user experience with minimal application sign-ons	
Customer wants to modernize branches with SASE	

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled — explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

CISCO PRODUCT / FEATURE	WHAT IT DOES / HOW IT WORKS	WHY YOU CHOSE IT (CISCO DIFFERENTIATOR)
Secure Access, Secure Client, Duo — ZTA module in Secure Client		
Secure Client VPN module (Full Tunnel)		
ZTA module, Secure Client		
Secure Access Private Resources proxy		
Duo		
Secure Access VPNaaS		

Secure Access Unified Policy		
Secure Access + SD-WAN		

PROCTOR USE ONLY — SCORING & FEEDBACK

CRITERION	MAX	SCORE
Pains & requirements mapping	40	
Cisco products (how & why)	30	
Diagram	30	
Total	**100**	

Overall level

- ☐ Associate 60–74
- ☐ Professional 75–89
- ☐ Expert 90–100

What the team did well

Where they can improve

Use Case #3 — AI Access and Agent Controls

Foundation · Required

100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here — or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

CUSTOMER PAIN POINT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Shadow AI usage is an issue today	
Over-privileged MCP servers for applications in the private and public DCs	
Users using personal credentials for ChatGPT login	
Downloading of risky AI models	

STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

CUSTOMER REQUIREMENT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Users should not be granted rights to use tools equally. Fine grained control is needed	
Corporate users should only be able to login to their organization's ChatGPT tenant	
Some remote users will be running Claude Desktop agents. Ensure your design includes controls for those employee systems.	
Some users will be integrating MCP servers to local agents like Claude Code	

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled — explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

CISCO PRODUCT / FEATURE	WHAT IT DOES / HOW IT WORKS	WHY YOU CHOSE IT (CISCO DIFFERENTIATOR)
AI Access within Secure Access		
AI Gateway within Secure Access		
Secure Client with agentic client discovery		
Duo IdP		

PROCTOR USE ONLY — SCORING & FEEDBACK

CRITERION	MAX	SCORE
Pains & requirements mapping	40	
Cisco products (how & why)	30	
Diagram	30	
Total	**100**	

Overall level

- ☐ Associate 60–74
- ☐ Professional 75–89
- ☐ Expert 90–100

What the team did well

Where they can improve

Use Case #4 — Merger / Acquisition; Extend Segmentation

Bonus · If time permits | 100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here — or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

CUSTOMER PAIN POINT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Unsure how to extend segmentation to acquired company	
Parent company not ready to add 100 newly acquired staff into their corporate directory. Is there a quick option of using what they already have?	
During transition, minimize disruptive user authentication experience to corporate apps and quickly secure acquired company apps	
Acquired company suffered identity breach previously. Still using passwords.	

STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

CUSTOMER REQUIREMENT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
CISO wants to extend segmentation to acquired company using TrustSec	
Only selected branch users can access a restricted resource at recently acquired company (Site 9951)	

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled — explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

CISCO PRODUCT / FEATURE	WHAT IT DOES / HOW IT WORKS	WHY YOU CHOSE IT (CISCO DIFFERENTIATOR)
FMC / FTD		
Duo — IdP (Directory)		
Duo — IdP (SSO routing)		
SCC		

PROCTOR USE ONLY — SCORING & FEEDBACK

CRITERION	MAX	SCORE
Pains & requirements mapping	40	
Cisco products (how & why)	30	
Diagram	30	
Total	**100**	

Overall level

- ☐ Associate 60–74
- ☐ Professional 75–89
- ☐ Expert 90–100

What the team did well

Where they can improve

Use Case #5 — Zero Trust across multicloud Data Centers

Bonus · If time permits | 100 pts

Sketch your proposed design, then explain how it solves each pain point and meets each requirement, and which Cisco products you would use and why.

STEP 1 · YOUR PROPOSED DIAGRAM

Sketch your proposed network diagram here — or paste one made with the Cisco Circuit app / an AI tool.

STEP 2 · ADDRESS THE CUSTOMER'S PAIN POINTS

CUSTOMER PAIN POINT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Minimal visibility of applications across multicloud data centers	
Lacking segmentation strategy and plan	
Wide deployment of Kubernetes with latency and visibility issues	
Unable to patch IoT devices	
Too many disparate security management interfaces for on-prem and multicloud enforcement points	

STEP 3 · MEET THE CUSTOMER'S REQUIREMENTS

CUSTOMER REQUIREMENT	YOUR TEAM'S RESPONSE — HOW YOUR CISCO DESIGN SOLVES THIS
Limited budget. Want to leverage native security features when practical. Open to virtual appliances	

CIO seeking end-to-end segmentation solution	
Minimize complexity	

STEP 4 · YOUR CISCO PRODUCTS

Suggested Cisco products are pre-filled — explain **how each works** and **why you'd choose it** (the differentiator). Use the blank row for anything else you'd add.

CISCO PRODUCT / FEATURE	WHAT IT DOES / HOW IT WORKS	WHY YOU CHOSE IT (CISCO DIFFERENTIATOR)
Secure Workload		
NGFW / FTD		
Isovalent		

PROCTOR USE ONLY — SCORING & FEEDBACK

CRITERION	MAX	SCORE
Pains & requirements mapping	40	
Cisco products (how & why)	30	
Diagram	30	
Total	**100**	

Overall level

- ☐ Associate `60–74`
- ☐ Professional `75–89`
- ☐ Expert `90–100`

What the team did well

Where they can improve

Glossary

ZTNA Zero Trust Network Access
Per-application access that replaces broad, over-permissive VPN.

SASE Secure Access Service Edge
Cloud-delivered networking and security combined.

SSE Security Service Edge
The security half of SASE (SWG, ZTNA, CASB, DLP…).

SGT Security Group Tag
An identity label used to segment traffic without VLANs or IP subnets.

TrustSec Cisco TrustSec
Cisco's SGT-based segmentation technology.

ISE Identity Services Engine
Authenticates and profiles users/devices and assigns SGTs.

SWG Secure Web Gateway
Inspects and filters internet-bound web traffic.

IdP Identity Provider
The system that authenticates users (e.g., Duo, Okta, Entra).

SSO Single Sign-On
One login for many applications.

MFA Multi-Factor Authentication
A second proof of identity beyond a password.

802.1X IEEE 802.1X
Port-based network access authentication.

MAB MAC Authentication Bypass
Authenticate a device by its MAC when it can't do 802.1X.

pxGrid Platform Exchange Grid
Cisco's framework for sharing context between security products.

SGACL Security Group ACL
Access rules written in terms of SGTs.

CSW Cisco Secure Workload
Application visibility and microsegmentation (formerly Tetration).

ADM Application Dependency Mapping
Discovers app traffic flows to build segmentation policy.

FTD / FMC Firewall Threat Defense / Firewall Mgmt Center
Cisco Secure Firewall and its manager.

NGFW / NGIPS Next-Gen Firewall / IPS
Modern firewall and intrusion-prevention.

MCP Model Context Protocol
How AI agents connect to external tools and data.

DCR Dynamic Client Registration
Registering and authorizing agents/apps with an IdP.

DLP Data Loss Prevention
Stops sensitive data from leaving the organization.

SCIM System for Cross-domain Identity Mgmt
Syncs users between directories (e.g., Okta → Duo).

VPNaaS VPN as a Service
Cloud-delivered VPN termination.

eBPF extended Berkeley Packet Filter
Kernel technology for observability and security (Cilium).

CNI Container Network Interface
Kubernetes networking plugin (e.g., Cilium).

SCC Security Cloud Control
Unified cloud management for Cisco security policy.

IoT Internet of Things
Networked devices like printers, cameras and badge readers.